DINGHYGO

the inflatable sailboat

Owner’s Manual

Section 4 — Assembly and disassembly

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2024

4. Assembly and disassembly

> 📌 **NOTE**
> Look closely at the DinghyGo instructional videos on assembling and disassembling the watercraft. You will receive these instruction videos with your DinghyGo. You can also order these from your DinghyGo supplier.

4.1 Inflatable floor

The DinghyGo boats are equipped with an inflatable floor that needs to be filled properly. Using the boat without a properly inflated bottom is unsafe, uncomfortable and can lead to damage to the boat. See section 4.7 to learn how to fill the inflatable floor.

4.2 Valves

The boat valves are designed especially for safe and comfortable use. The flat valves are designed to increase the comfort in the boat and to prevent damage to the boat.

The valve is closed when the middle screw knob is on top. The outer cap protects the valve from dirt when not in use.

Fig. 4.1 — Valve

4.2.1 Using the valves

- Remove the outer cap. The valve is closed when the middle screw knob is on top.
- To open the valve: put your finger on the centre of the valve and press down the centre screw button. Turn your finger ¼ clockwise until the knob locks.
- To close the valve: press the knob and turn your finger ¼ clockwise until the knob comes up.

4.2.2 Pump connections

- Attach the nozzle of the pump (the tip part) to the valve.
- Turn it to the right (clockwise) until it locates, and start pumping.
- Keep pumping until the boat is completely full.
- Unplug the pump when finished.
- Make sure the cap is put back on (to protect against dirt and damage).

4.2.3 Checking for leaks

If air escapes:

- Take the valve key out of the repair kit.
- Insert the key into the valve and turn the key clockwise. Check if the problem is resolved.

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- When a leak is discovered, contact your DinghyGo supplier and check the warranty.
- Take the back of the valve into your hands and turn the stem to the left with the key (counter clockwise) and remove the stem.
- Inspect the valve for damage.
- If there is any damage (see Appendix 8: Warranty), take the defective valve to your DinghyGo supplier. You will receive a new valve within the warranty period.
- Lubricate the stem with silicone or soapy water and reinstall it.

4.3 Rowing kit

There are two oars supplied with the boat. The blades can be removed from the shafts to facilitate transportation. The oars should be placed in the oarlocks.

4.4 Sail kit

Your DinghyGo is supplied with a sail kit, consisting of:

- Seat / daggerboard case
- Mast step board
- Daggerboard
- Rudder
- Sectional mast
- Sectional boom
- Sail
- Rope set *(Orca 4.8 and 5.6 m² rig includes additional supporting shrouds with cleats and shackle)*
- Sail kit bag

The installation and rigging of the sailing equipment is explained in section 4.8 and in Appendix 9 (Rudder assembly) and Appendix 10 (Orca boom assembly).

4.5 Seating

DinghyGo is equipped with a special seat and a "seat cord" on both side tubes, onto which the seat is pushed. The seat also functions as the top slot for the daggerboard case.

> 📌 **NOTE**
> You must install the seat **before** fully inflating the boat.

4.6 Inflating the tubes

Roll out the boat onto the ground to fill it up with air. Remove any sharp objects from the surface on which the boat is inflated.

- After unpacking the boat, check if all the components are present.

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- Check that the valves are closed. This is done by putting a finger in the valve, pushing the little grey screw button and turning it slightly to the right.
- If the knob jumps up a little, you can proceed to fill the boat with air. *(To drain air: turn the knob to the left.)*
- Use the pump provided with the boat. Push the nozzle against the valve and turn to the right — the pump will lock onto the valve.
- Put enough air in the boat to create the correct shape.

> 📌 **NOTE**
> All inflatable chambers must be inflated **equally** to avoid damage to the bulkheads.

> 📌 **NOTE**
> Inflating and deflating your boat the right way is essential to the durability of your boat.

4.6.1 Order in which to inflate

Inflate the boat in the ascending order of the numbers on the valve tags:

1. Front chamber (no. 1)
2. Side-chambers (no. 2 & 3)
3. Floor (no. 4)

4.6.2 Inflating the hull tubes

Step 1 Pump air into the front chamber using the inflation valve until the specified pressure is reached.

Step 2 Pump air into the two side-chambers in the correct order until the specified pressure is reached. If inflation is carried out in the correct order, the boat will have the correct pressure and there will not be any deformation of the float sections.

Step 3 Seal all the valves by inserting the valve caps and turning them clockwise.

> 📌 **NOTE**
> Never exceed the values on the builder's plate and valve tags!
> Inflate the boat floats to **0.25 bar / 3.6 psi / 25 kPa** and the floor to **0.19 bar / 2.75 psi / 19 kPa** (tolerance ±20%).
> Inflate the Orca floor to **0.7 bar / 10.2 psi / 70 kPa**.

> 📌 **NOTE**
> Do not use mechanical compressors to inflate your boat. The included hand pump is adjusted to create the perfect pressure. If preferred, you could use an electric pump available as an option from your DinghyGo supplier.

4.7 Inflating the inflatable floor

If you fully inflate the inflatable floor, it clamps in between the side tubes and the transom.

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Step 1 Pump air into the bottom through the valve, like the side tubes, until the specified pressure is reached.

Step 2 Seal the valve by inserting the valve cap and rotating clockwise.

4.8 Assembly and disassembly of the sailing rig

> ⚠️ **CAUTION**
> Incorrect inflation can cause structural damage to your watercraft. Take your watercraft out of direct sunlight when outside of the water. Direct sunlight causes the air in the watercraft to expand, causing damage to your watercraft (except through proper use and maintenance of the pressure relief valve).

The sail kit bag contains the complete sailing equipment. Lay out all sail components to facilitate installation and rigging.

4.8.1 Installation of the mast step board, seat and bridle

- Partially inflate the bow and side tubes (see section 4.6.2), so that they have some shape but are not fully up to pressure. Do not inflate the floor yet.
- Slide the mast step board in under the tubes until it locates in front of the mast step board stoppers on the floor (at front of boat).
- Slide the seat / daggerboard case onto the seat cords on the tubes, until it is located directly above the daggerboard case opening in the bottom.
- Tie one end of the bridle onto the transom eye and attach the other end to the other transom eye with the carabiner.

4.8.2 Inflation of the boat

- Inflate the hull tubes and floor completely (see sections 4.6.2 and 4.7), so that the mast step board and seat / daggerboard case are firmly clamped.
- Attach the flexible part of the daggerboard case with the Velcro on the bottom of the seat / daggerboard case.
- Attach the oars (see section 4.9).

4.8.3 Rigging

- Make sure the rudder and Orca boom are correctly assembled after unboxing (see Appendix 9 and Appendix 10).
- Slide the mast and boom sections together.
- Take the sail and insert the sail battens.
- Slide the mast into the mast sleeve of the sail.
- *(Orca 4.8 and 5.6 m² rig only)* Attach the supporting shrouds through the mast sleeve opening with the shroud shackle to the top mast eye.
- Place the mast in the mast step through the deck hole, and lock the mast under the deck with the push-button system *(Nomad only)*.
- *(Orca 4.8 and 5.6 m² rig only)* Attach the supporting shrouds to both sides of the tubes by leading the shroud through the D-ring and securing it in the shroud cleat. Make sure the mast stands straight with just a little shroud tension.

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> ⚠️ **WARNING**
> Make sure the sailing rig is properly inserted into the mast step and that the mast is locked underneath the deck with the push-button system. Improper attachment of the sailing rig can lead to the loosening of the rig, causing the watercraft to get out of control and potentially injuring the crew.

> ⚠️ **WARNING**
> Make sure the mast stands straight and is properly supported by the shrouds *(Orca 4.8 and 5.6 m² rig)*. Improper attachment of the supporting shrouds can lead to loss of control of the watercraft and injury to the crew.

- Attach the boom to the mast by snapping the front end of the boom to the mast.
- Attach both Velcro straps from the sail to the boom.
- Apply the downhaul and outhaul with the correct tension.
- Attach the mainsheet block with snap shackle to the bridle.

4.8.4 Preparation before going into the water

- Attach the rudder to the transom. Ensure that the rudder is locked with the rudder retaining clip.

> ⚠️ **WARNING**
> Attach the rudder with the retaining clip. Incorrect securing can lead to loss of the rudder and consequent loss of direction of the watercraft, which can lead to life-threatening situations.

- Slide the daggerboard through the seat into the daggerboard case.
- Close the self-bailer.

> 📌 **NOTE**
> The daggerboard has to be inserted into the daggerboard case with the **blunt** profile aimed to the bow.

> 📌 **NOTE**
> The bridle has to be attached **over** the tiller of the rudder.

> ⚠️ **WARNING**
> Always secure the tiller in the rudder head before sailing. The rudder can otherwise be damaged, leading to loss of direction which could result in life-threatening situations.

4.8.5 Sailing

> ⚠️ **WARNING**
> Only sail once you and your crew have adequate sailing experience. Improper use, faulty equipment and unsuitable conditions can cause life-threatening situations.

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This manual does not provide sailing instructions and assumes that the operator and passengers have adequate sailing experience and qualifications. Your DinghyGo supplier or sailing club is able to inform you of training opportunities.

4.8.6 Disassembly

- Take the rudder off of the transom by pressing/releasing the rudder retaining clip.
- Remove the daggerboard from the daggerboard case and seat.
- Disconnect the mainsheet block from the bridle by opening the snap shackle.
- *(Orca 4.8 / 5.6 m² only)* Detach the supporting shrouds from both sides of the tube D-rings.
- Raise the sailing rig out of the mast step through the deck, after pressing the push-button system on the mast below deck.
- Release the outhaul and downhaul and detach the boom from the mast.
- Take the boom sections apart.
- *(Orca 4.8 and 5.6 m² rig only)* Remove the battens from the sail, detach the supporting shrouds from the mast and pull the mast sleeve off of the mast.
- Take the mast sections apart.

4.8.7 Removal of the mast step board and seat

- Partially deflate the floor and side tubes (see section 4.11), so that the boat still has shape but is not pressurised.
- Detach the flexible portion of the daggerboard case from the underside of the seat.
- Slide the seat backward from its mounting tendons on the floats.
- Pull the mast step board out of the stoppers from the bow. Lifting the bow with one hand makes it easier to take out.
- Take out the oars from the oarlocks and take apart the oars.

4.8.8 Packing up and storing the sail kit

- Ensure that all sail kit parts are clean and dry. Parts subjected to salt water should be washed with fresh water immediately after use.
- Lay out the sail kit bag on the ground with the opening facing towards you.
- Put the daggerboard and rudder in the bag first, then stack the mast and boom sections on top.
- Add the loose oar components, seat, mast step board, folded sail, battens and rope set.
- Finally, add the hand pump.
- Shut the bag firmly with the straps and buckles. Carry and store in a sheltered, dry place.

> 📌 **NOTE**
> When packing, the outhaul, downhaul and mainsheet ropes and blocks remain attached to the boom. The bridle remains attached to the transom eyes in the boat.

4.9 Installation of the oars

The DinghyGo boats are equipped with divisible oars, oarlocks and a combined rowing seat and daggerboard case.

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> ⚠️ **CAUTION**
> Incorrect attachment of the oars can damage the attachment pins and oarlocks. Remove superfluous plastic from the bottom of the oar lock opening when necessary.

- Make sure the seat is installed correctly (see section 4.8.1).
- Hold the oars parallel to the tubes with the blade facing forwards *(not backwards — they won't go in)* and insert the metal fastening pin into the oarlock as deep as possible.
- Turn the oars with the blades outward, so they are secured in the oarlock and can be used correctly.
- When not in use, oars can be stored along the tubes, secured by the oarlocks and the elasticated oar holders at the aft end.

> 📌 **NOTE**
> Take local conditions into account before heading out. It is possible that the propulsive power of the boat is not strong enough to sail against a strong tide or river flow.

4.10 Installation of an outboard motor

> 📌 **NOTE**
> Consult the Model specification table for maximum outboard motor power and weight.

> 📌 **NOTE**
> Using an outboard motor that exceeds the maximum horsepower or weight may result in:
> - Difficulties with manoeuvrability and/or stability
> - Changes to the designed flotation or sailing characteristics
> - Damage to the boat, in particular the areas around the transom

> 📌 **NOTE**
> Use an emergency stop switch. This will stop the outboard motor if the operator has left the controls.

4.10.1 Mounting the outboard motor

- Make sure you are standing in a safe and stable position.
- Unlock the outboard motor so that it becomes possible to tilt the clamp, then mount it onto the transom using the clamp screws.
- The outboard motor must be in the middle of the transom. When lowering the outboard onto the transom, take care not to damage the rudder gudgeons.
- The clamp bracket should be screwed securely onto the motor plates. Check this periodically as clamp screws can vibrate loose.

4.10.2 Position of the outboard motor

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⛔ **DANGER**
Using an outboard motor with too much power or weight can result in serious injury, death or damage to the watercraft.

⚠️ **CAUTION**
Some outboard motors may damage the rudder gudgeons when being attached to the transom. Make sure your outboard motor fits and, if necessary, have it adjusted by an expert or ask your DinghyGo supplier for help.

The outboard motor has to be installed in such a way that it stands vertically in the water when in the normal operating position. The "cavitation plate" at the bottom should be situated horizontally in the water when the boat is in its normal position.

4.10.3 Starting the outboard motor

- Pull the boat into the water.
- Secure the outboard motor in the correct position with the clamp screws.
- Make sure you are standing securely and then start the outboard motor.
- Avoid high speeds in reverse — water can enter the boat over the transom.

📌 **NOTE**
Ensure that the self-bailer in the transom is **closed** before taking the boat into the water.

4.11 Deflation of the boat

📌 **NOTE**
Make sure the boat is clean and dry before rolling it up for storage. Remove any sand and debris from the skin material.

📌 **NOTE**
Do not let the boat deflate chamber by chamber. Deflate the chambers **evenly** — this prevents damage to the bulkheads.

Deflation procedure:

- Put the boat on the ground.
- Put your finger in the valve and turn the grey screw button to the right (see section 4.2 for more information on valves).
- Let some air out of the hull tubes and floor.
- Make sure the inside of the valve shrouds open to enable air to escape during folding.
- Push evenly onto the whole boat to expel as much air as possible.

4.12 Packing up and storing the boat

- Put the boat on the ground and remove all sailing and rowing parts, including seat, mast step board and oars.

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- Fold the side tubes toward the middle, so that the whole boat is the same width as the transom.
- Fold the bow over the deck.
- Pull the bow and deck backwards towards the end of the daggerboard case.
- Fold the round-shaped aft ends of the tubes towards the middle of the transom.
- Lift and pull the transom forwards over the already folded portion.
- Place the compact folded package onto the spread-out boat bag.
- Fold the two small, then the two large bag pieces around the package and fasten with the straps and buckles.

4.13 Transport

4.13.1 Getting the boat out of the water

Make sure that there are no sharp edges at the point where you want to lift the boat out of the water. Preferably use the lifting eyes or handles for lifting, not the ropes.

4.13.2 Towing the boat

📌 **NOTE**
If the inflatable boat has to be towed by another boat, it must be **empty**:
- Remove the sailing rig, outboard motor, fuel tank, batteries and other equipment.
- Close off the daggerboard case with the provided flexible insert (black EVA material), roll up the flexible part of the daggerboard case, and buckle up before closing.

! IMPORTANT
The handle on the bow must **not** be used for towing, anchoring or mooring.

- Use the metal rings on the port and starboard sides of the bow for towing. This allows the boat to remain stable behind the towing vessel.
- Attach a line between the towing rings to form a cross. Attach a towing line to this and tow the boat at low speed.

⚠️ **WARNING**
- Never tow the watercraft with people on board.
- Check the towing line repeatedly.
- Periodically check the towing conditions and make sure no water enters the watercraft.

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3. Specifications, description and functions

3.1 Specifications

model	Orca 280	Orca 325	Orca 375
Boat specifications			
CE category	D	D	D
Number of air chambers	3	3	3
Dimensions and weight *			
length (ft/cm)	9'2"/280	10'7"/325	12'3"/375
width (ft/cm)	4'9"/150	4'9"/150	4'9"/150
draft (ft/cm)	2'8"/85	2'8"/85	2'8"/85
tube diameter (in/cm)	16"/40	16"/40	16"/40
weight (lbs/kg)	60/27	64/29	71/32
Capacity (maximum)			
persons (without/with rigging)	3/2	4/2	5/2,5
payload (lbs/kg)	300	1,265/575	1,430/650
outboard engine power (kW/Hp)	4.5/6.0	4.5/6.0	4.5/6.0
outboard engine weight (lbs/kg)	66/30	66/30	66/30
pressure tubes/floor (psi)	3.6/10.2	3.6/10.2	3.6/10.2
Packing size ** (LxWxH in)	40"×21"×13"	40"×28"×13"	40"×28"×15"
Packing size ** (LxWxH cm)	100×54×32	100×72×32	100×72×38
Sail/row-kit specifications			
*Dimensions and weight **			
mast length (ft/cm)	11'3"/345	14'1"/430	14'1"/430
boom length (ft/cm)	6'7"/205	7'4"/225	7'4"/225
sail area (sq.ft/m²)	42/3.9	60/5.6	64/5.9
weight (lbs/kg)	31/14	33/15	37/17
Packing size ** (LxWxH in)	46"×18"×6"	47"×18"×7"	47"×18"×8"
Packing size ** (LxWxH cm)	118×45×16	120×45×18	120×45×20

* Specifications can differ by 5%
** Weights and optimal packing sizes excluding carton, packing materials & bag.

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2. Specifications, description and functions

2.7 Visual representations

Sideway view — Orca 375

Additional items — shrouds mast support *(main sail > 3.9 m²)*

Additional items — jib

66. Halyard

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60. Mast sleeve opening
61. Mast eyelet
62. Funny shackle
63. Funny
64. Funny cleat
65. Funny/davits D-ring
67. Halyard shackle
68. Jib head
69. Jib
70. Halyard cleat
71. Jib tack
72. Jib shackle (clew)
73. Bow D-ring
74. Jib clew
75. Jib sheet
76. Jib sheet D-ring

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